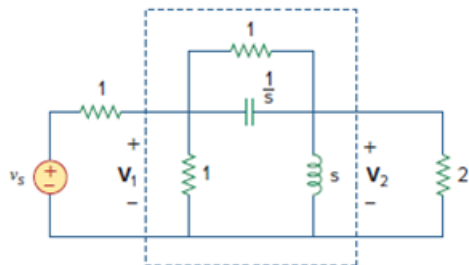


## Problem

(a) Find the  $y$  parameters of the two-port in Fig. 19.84.

(b) Determine  $\mathbf{V}_2(s)$  for  $v_s = 2u(t)$  V.

**Figure 19.84**



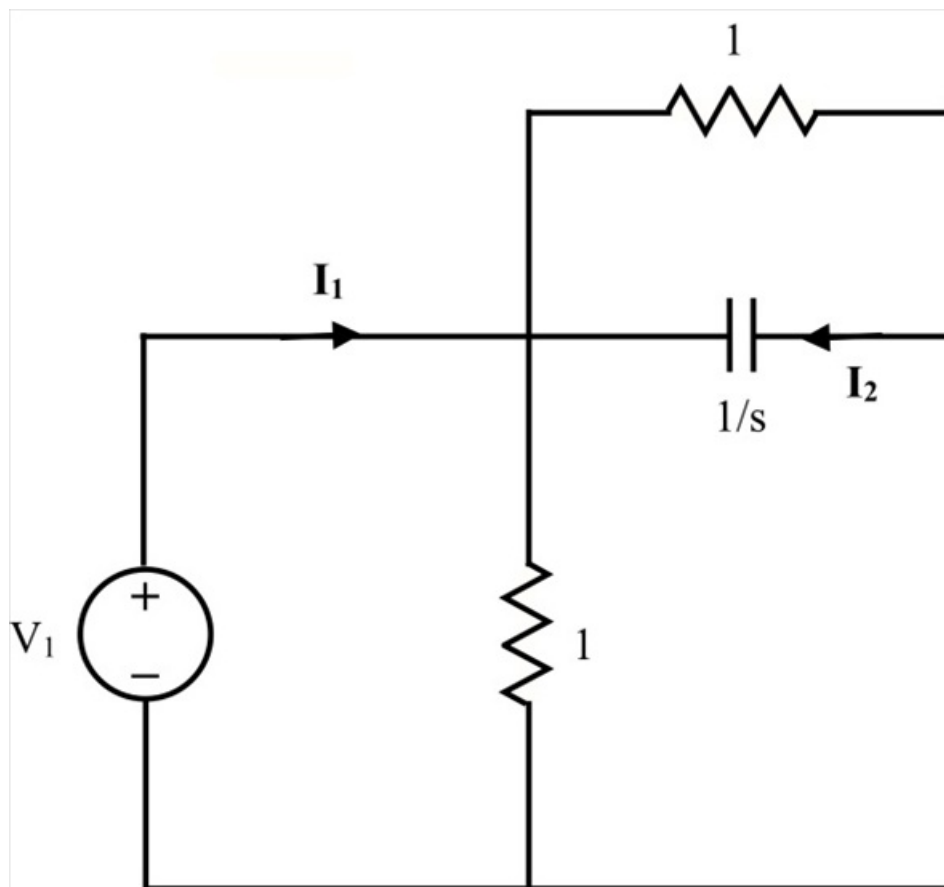
## Step-by-step solution

## Step 1 of 10

(a)

Refer to Figure 19.84 in the textbook.

Draw the circuit for  $\mathbf{V}_2 = 0$  as shown in Figure 1.



**Figure 1**

## Step 2 of 10

Determine the value of  $Y_{11}$ .

$$Y_{11} = \frac{I_1}{V_1} \Big|_{V_2=0}$$

$$Y_{11} = \frac{V_1}{(1 \parallel Y_s) \parallel 1}$$

$$= \frac{1}{\left(\frac{1}{s+1}\right) \parallel 1}$$

$$= \frac{1}{\left[\frac{\left(\frac{1}{s+1}\right) 1}{\frac{1}{s+1} + 1}\right]}$$

$$= \frac{1}{\left[\frac{1}{s+2}\right]}$$

$$Y_{11} = s + 2$$

### Step 3 of 10

Calculate the value of  $Y_{21}$ .

$$Y_{21} = \frac{I_2}{V_1} \Big|_{V_2=0}$$

$$Y_{21} = \frac{I_2}{V_1} \Big|_{V_2=0}$$

$$I_2 = -\frac{V_1}{\left(\frac{1}{s+1}\right)}$$

$$\frac{I_2}{V_1} = -(s+1)$$

$$Y_{21} = -(s+1)$$

### Step 4 of 10

Draw the circuit for  $V_1 = 0$  as shown in Figure 2.

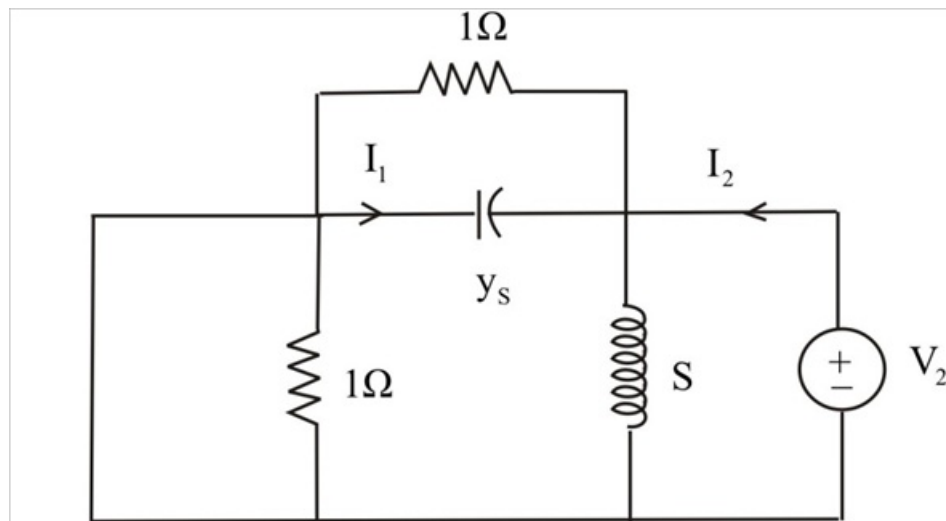


Figure 2

Step 5 of 10

Determine the value of  $Y_{22}$ .

$$Y_{22} = \frac{I_2}{V_2} \Big|_{V_1=0}$$

$$I_2 = \frac{V_2}{\left( \left( \frac{1}{s} \parallel 1 \right) \parallel s \right)}$$

$$= \frac{V_2}{\left( \frac{\left( \frac{1}{s+1} \right)s}{s + \frac{1}{s+1}} \right)}$$

Step 6 of 10

Consider,

$$I_2 = \frac{V_2}{\left( \frac{s}{s^2 + s + 1} \right)}$$

$$Y_{22} = \frac{s^2 + s + 1}{s}$$

Calculate the value of  $Y_{12}$ .

$$Y_{12} = Y_{21}$$

$$= -(s+1)$$

Therefore, the y parameters of the network is

$$Y = \begin{bmatrix} s+2 & -(s+1) \\ -(s+1) & \frac{s^2 + s + 1}{s} \end{bmatrix}$$

Step 7 of 10

(b)

Draw the modified circuit as shown in Figure 3.

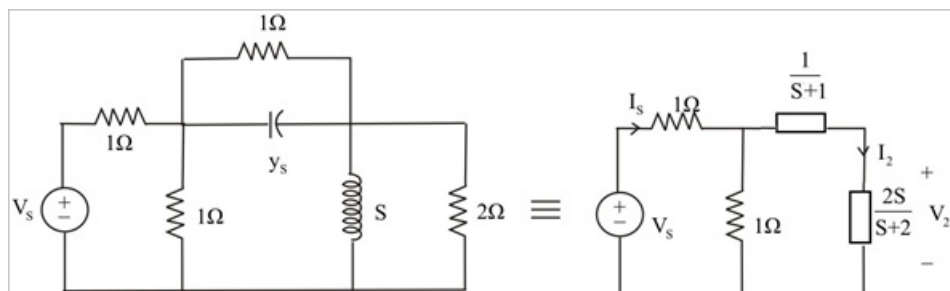


Figure 3

Step 8 of 10

Write the equation for source current  $\mathbf{I_s}$ .

Consider  $\mathbf{V_s} = \frac{2}{s}$ .

$$\begin{aligned}\mathbf{I_s} &= \frac{\mathbf{V_s}}{\left[ \left( \frac{2s}{s+2} + \frac{1}{s+1} \right) \parallel 1 \right] + 1} \\ &= \frac{\frac{2}{s}}{\left\{ \left[ \frac{\frac{2s^2 + 2s + s + 2}{(s+1)(s+2)} \times 1}{\frac{2s^2 + 2s + s + 2}{(s+1)(s+2)} + 1} \right] + 1 \right\}} \\ &= \frac{\frac{2}{s}}{\frac{(2s^2 + 3s + 2)}{3s^2 + 6s + 4} + 1} \\ \mathbf{I_s} &= \frac{\frac{2}{s}}{\left( \frac{2s^2 + 3s + 2 + 3s^2 + 6s + 4}{3s^2 + 6s + 4} \right)} \\ &= \frac{2(3s^2 + 6s + 4)}{S(5s^2 + 9s + 6)}\end{aligned}$$

**Step 9** of 10

Calculate the current  $\mathbf{I_2}$ .

$$\begin{aligned}\mathbf{I_2} &= \frac{\mathbf{I_s}}{\left( 1 + \frac{1}{s+1} + \frac{2s}{s+2} \right)} \\ &= \frac{\mathbf{I_s}(s+1)(s+2)}{s^2 + 3s + 2 + s + 2 + 2s^2 + 2s} \\ &= \frac{\mathbf{I_s}(s+1)(s+2)}{3s^2 + 6s + 4} \\ &= \frac{2(s+1)(s+2)}{S(5s^2 + 9s + 6)}\end{aligned}$$

**Step 10** of 10

Determine the voltage  $\mathbf{V_2(s)}$ .

$$\begin{aligned}\mathbf{V_2} &= \mathbf{I_2} \times \frac{2s}{s+2} \\ &= \frac{2(s+1)(s+2)}{S(5s^2 + 9s + 6)} \times \frac{2s}{s+2} \\ &= \frac{2(s+1) \times 2}{(5s^2 + 9s + 6)} \\ &= \frac{4(s+1)}{5s^2 + 9s + 6} \\ &= \frac{08(s+1)}{s^2 + 1.8s + 1.2}\end{aligned}$$

Therefore, the voltage  $\mathbf{V_2(s)}$  is  $\boxed{V_2(s) = \frac{08(s+1)}{s^2 + 1.8s + 1.2}}$ .

